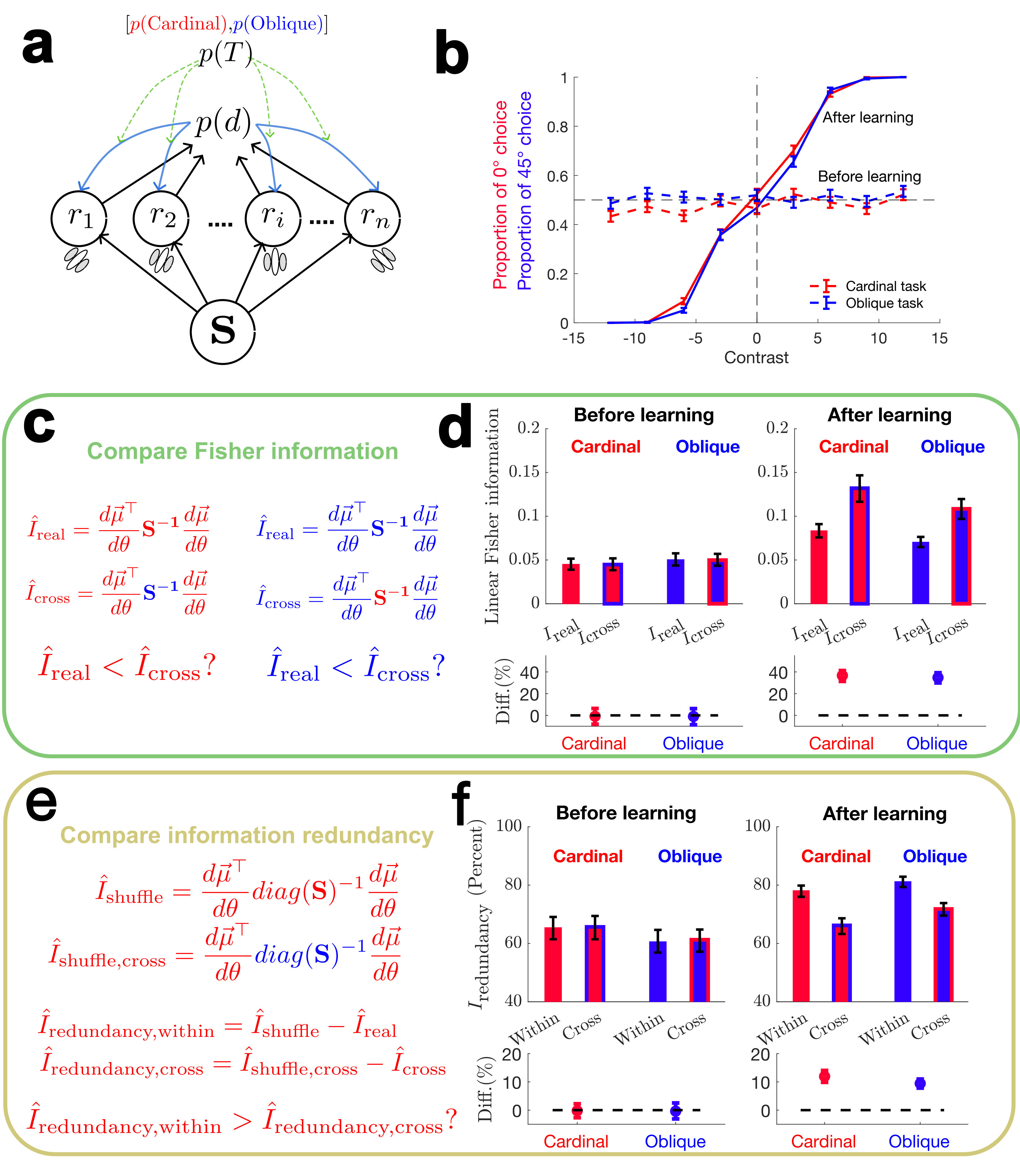

a
[p(Cardinal),p(Oblique)]
p(T)
p(d)
r1
r2
....
ri
....
rn
S
b
Proportion of 0° choice
Proportion of 45° choice
After learning
Before learning
Cardinal task
Oblique task
Contrast
1
0.8
0.6
0.4
0.2
0
-15
-10
-5
0
5
10
15
c
Compare Fisher information
$\hat{I}_{\text{real}} = \frac{d\vec{\mu}^{\top}}{d\theta}\mathbf{S}^{-1}\frac{d\vec{\mu}}{d\theta}$
$\hat{I}_{\text{real}} = \frac{d\vec{\mu}^{\top}}{d\theta}\mathbf{S}^{-1}\frac{d\vec{\mu}}{d\theta}$
$\hat{I}_{\text{cross}} = \frac{d\vec{\mu}^{\top}}{d\theta}\mathbf{S}^{-1}\frac{d\vec{\mu}}{d\theta}$
$\hat{I}_{\text{cross}} = \frac{d\vec{\mu}^{\top}}{d\theta}\mathbf{S}^{-1}\frac{d\vec{\mu}}{d\theta}$
$\hat{I}_{\text{real}} < \hat{I}_{\text{cross}}$?
$\hat{I}_{\text{real}} < \hat{I}_{\text{cross}}$?
d
Before learning
After learning
Cardinal
Oblique
Linear Fisher information
0.2
0.15
0.1
0.05
0
Ireal
Icross
Diff.(%)
40
20
0
Cardinal
Oblique
e
Compare information redundancy
$\hat{I}_{\text{shuffle}} = \frac{d\vec{\mu}^{\top}}{d\theta} diag(\mathbf{S})^{-1}\frac{d\vec{\mu}}{d\theta}$
$\hat{I}_{\text{shuffle,cross}} = \frac{d\vec{\mu}^{\top}}{d\theta} diag(\mathbf{S})^{-1}\frac{d\vec{\mu}}{d\theta}$
$\hat{I}_{\text{redundancy,within}} = \hat{I}_{\text{shuffle}} - \hat{I}_{\text{real}}$
$\hat{I}_{\text{redundancy,cross}} = \hat{I}_{\text{shuffle,cross}} - \hat{I}_{\text{cross}}$
$\hat{I}_{\text{redundancy,within}} > \hat{I}_{\text{redundancy,cross}}$?
f
Before learning
After learning
Cardinal
Oblique
$I_{\text{redundancy}}$ (Percent)
100
80
60
40
Within
Cross
Diff.(%)
20
10
0
Cardinal
Oblique